

What is DNS?

DNS stands for **Domain Name System**, and it's often called the "phonebook of the internet".

When you type a website address like google.com into your browser, your computer doesn't understand words – it understands numbers called **IP addresses** (for example, 142.250.190.46). DNS is the system that translates the human-friendly name (google.com) into the machine-friendly number (the IP address) so your browser can load the website.

Without DNS, you'd have to memorise a long string of numbers for every site you visit – impossible in practice. DNS does that translation for you, silently, every time you click a link or type a URL.

What actually happens when you type a website address?

When you type shoaibnagy.netlify.app into your browser and press Enter, a whole chain of events takes place "behind the scenes". Here's the step-by-step, using the library analogy from Cloudflare:

1. **Your browser asks a "DNS resolver"**

The resolver is like a librarian. You ask the librarian to find the book (the website) you want. The resolver's job is to track down the answer on your behalf.

2. **The resolver asks the "root nameserver"**

The root server is like the library's main index. It doesn't know the exact IP address, but it knows which section of the library to look in – in our case, it points to the nameserver responsible for the .app part of the address.

3. **The resolver then asks the "TLD nameserver"**

TLD stands for **Top-Level Domain** – that's the last part of the name, like .com, .org, or .app. This server is like a specific rack of books in the library. It knows which nameserver holds the records for netlify.app itself.

4. **Finally, the resolver asks the "authoritative nameserver"**

This is the final stop – the dictionary on the rack that actually contains the definition. The authoritative nameserver for netlify.app has the exact IP address for shoaibnagy.netlify.app. It returns that IP address to the resolver.

5. **The resolver gives the IP address back to your browser**

Your browser now has the numeric address it needs. It connects to that IP, requests the website, and the page loads in your window.

All of this happens in a fraction of a second, and you never see the requests flying back and forth – it's completely transparent to you as a user.

What is a CNAME record?

A **CNAME** stands for “Canonical Name” – it's essentially an **alias** or a forwarding address for a domain.

Imagine a scavenger hunt where one clue points to another clue, and the final clue points to the treasure. A CNAME record works exactly like that: it points from one domain name to another domain name, and then the DNS system follows that chain until it finds a real IP address.

For example, if you own `blog.example.com` and you set a CNAME record that points to `example.com`, then when someone types `blog.example.com`, the DNS system will “follow” that alias to `example.com` and then look up the IP address of `example.com`.

Key rules to remember about CNAME records:

- A CNAME must **always** point to another domain name – **never** directly to an IP address.
 - You can chain CNAME records (one CNAME pointing to another CNAME), but that slows things down because it takes extra lookups.
 - CNAMEs are especially useful when you have many subdomains (like `blog.example.com`, `shop.example.com`) that all share the same IP address. If that IP address ever changes, you only need to update the main domain's record, and all the CNAME aliases automatically follow.
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Putting it all together

When you type `shoaibnagy.netlify.app`:

- The resolver, root server, TLD server, and authoritative server work together to find the IP address.

- If there were a CNAME record involved (say I owned shoaibnagy.com and pointed it to shoaibnagy.netlify.app), the resolver would follow that alias first, then continue the lookup.
- Once the IP address is found, your browser connects and loads the site.

That's DNS in a nutshell – a quiet, essential system that makes the internet usable for humans, and CNAME records are just one of the tools it uses to keep things flexible and manageable.